CHB-MIT EEG Dataset Description

Neural activity associated to different cognitive tasks can be observed and quantified by means of physiological measures. Electroencephalogram (EEG) is a widely used tool due to its non-invasive nature and its high temporal resolution. The goal is to characterize cognitive states by means of artificial neural networks for learning EEG signals. In particular, we aim at identifying epileptic seizures.

1. Raw Data

Data was acquired from the CHB-MIT Scalp EEG Database (https://physionet.org/content/chbmit/1.0.0/). This dataset contains EEG recordings from 23 pediatric and one adult patient with intractable seizures. Recordings were made for several hours per patient and records are delivered in one hour-long files. Records that contain seizures are labelled as seizure records; the rest of records are labelled as non-seizure records. As ground truth, the start and the end time (in seconds) for each seizure are provided, based on expert judgements. A seizure record can contain one epileptic seizure or more.

Most of the recordings in the dataset contain 23 channels (computed as the differences between electrodes) recorded by an EEG cap with 21 electrodes. Signals were collected at the sampling frequency of 256 Hz. Figure 1 shows the EEG montage, and the arrows illustrates the channel combinations (longitudinal-bipolar). Table 1 shows the channels computed as the differences between pairs of electrodes.

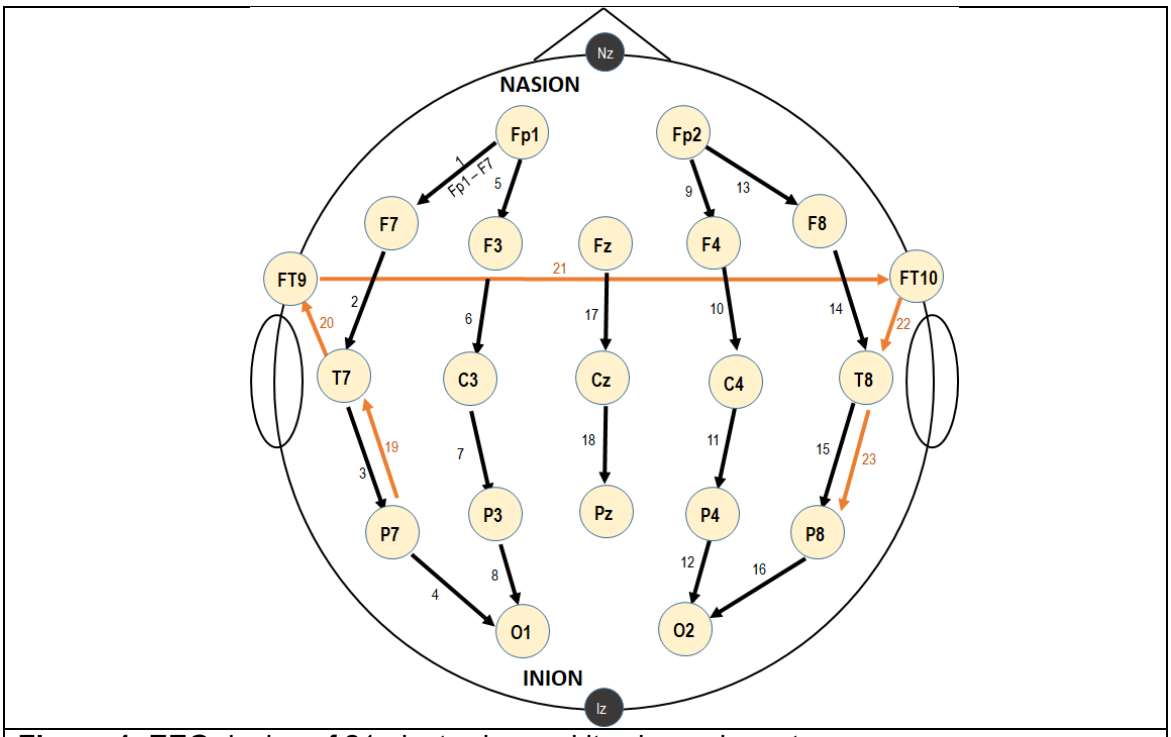

Figure 1. EEG device of 21 electrodes and its channel montage.

Table1: Channel names

Channel Id	Channel Name
Channel 1	FP1-F7
Channel 2	F7-T7
Channel 3	T7-P7
Channel 4	P7-O1
Channel 5	FP1-F3
Channel 6	F3-C3
Channel 7	C3-P3
Channel 8	P3-O1
Channel 9	FP2-F4
Channel 10	F4-C4
Channel 11	C4-P4
Channel 12	P4-O2
Channel 13	FP2-F8
Channel 14	F8-T8
Channel 15	T8-P8
Channel 16	P8-O2
Channel 17	FZ-CZ
Channel 18	CZ-PZ
Channel 19	T7-FT9
Channel 20	FT9-FT10
Channel 21	FT10-T8

For each patient, the dataset provides EEG recordings of many consecutive hours in EDF files (European Data Format), usually of one hour of duration. Files which have at least one seizure are annotate as seizure files, otherwise as non-seizure files. Besides, the dataset furnishes annotation of the time in which an epilepsy seizure starts and ends (this time is reference to the starting time of the seizure file). Table 2 shows patient ID and seizures during recordings.

Table 2: The dataset and patient information.

Patient	Sex – Age	Number of seizures	Total duration of seizures (sec)	Total recording time (h)	Total time of seizure files (h)
chb01	F-11	7	442	40.55	6.65
chb02	M-11	3	172	35.27	2.27
chb03	F-14	7	402	38.00	7.00
chb04	M-22	4	378	156.07	10.66
chb05	F-7	5	558	39.00	5.00
chb06	F-1.5	10	153	66.74	25.89
chb07	F-14.5	3	325	67.05	9.04
chb08	M-3.5	5	919	20.01	5.00

chb09	F-10	4	276	67.87	9.58
chb10	M-3	7	447	50.02	14.02
chb11	F-12	3	806	34.79	2.79
chb12	F-2	27	989	20.69	9.68
chb13	F-3	10	440	11.00	7.00
chb14	F-9	8	169	26.00	7.00
chb15	M-16	20	1992	39.01	14.01
chb16	F-7	8	69	17.00	5.00
chb17	F-12	3	293	20.01	3.01
chb18	F-18	6	317	34.63	5.63
chb19	F-19	3	236	28.93	2.93
chb20	F-6	8	294	27.60	5.57
chb21	F-13	4	199	32.83	3.83
chb22	F-9	3	204	31.00	3.00
chb23	F-6	7	424	26.56	8.96
chb24		16	511	21.30	12.00
Total		**181**	**11015**	**951.93**	**185.51**

Figure 2 illustrates a segment of EEG data from patient 1 with a seizure starting at the 2996 second highlighted in bold face. Although the shape of seizure can vary across patients due the type of epilepsy, individuals tend to keep similar patterns. For instance, it is noticeable the spatial and temporal consistency in the seizure of patient 1 shown in Fig.2. Figure 3 shows EEG signal from a single electrode with a seizure interval highlighted in red.

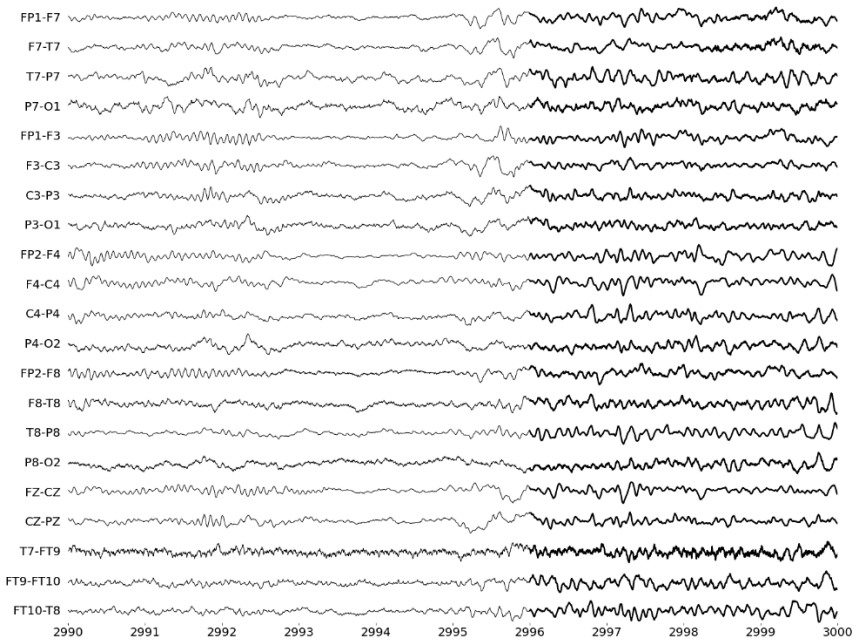

Figure 2. EEG signals differences between channels for a recording with a seizure (highlighted in bold) at the end of the recording.

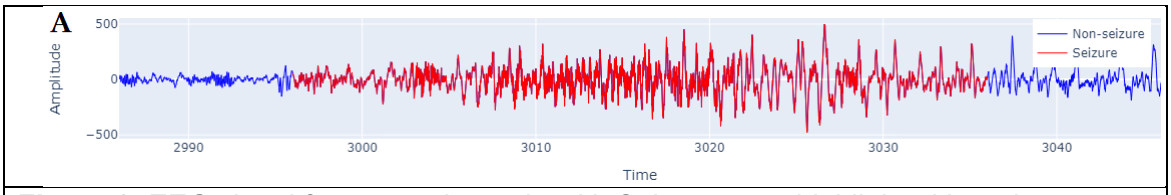

Figure 3. EEG signal from one electrode with Seizure state highlighted in red.

2. Input Data:

The original MIT data files have been pre-processed to provide EEG signals for each case in parquet format and annotations of each of the recordings

For each patient, EEG recordings have been processed and stored in different formats. The original .edf recordings have been converted into .parquet files, containing both seizure and non-seizure intervals, based on annotations. These files include recordings with 23 and 18 channels, while records with different sensor and channel distributions remain in the original EDF files. Data has been down-sampled to 128 Hz to optimize storage while maintaining sufficient resolution for seizure detection. Additionally, EEG signals have been segmented into 1-second temporal windows. To avoid transitional patterns near seizures, windows within ±30 seconds before or after a seizure were discarded. As illustrated in Figure 1, the data is categorized into normal (non-seizure), preictal (before seizure), ictal (seizure), and postictal (after seizure) intervals. Non-ictal windows are non-overlapping, whereas ictal windows have an 80% overlap for data augmentation. These windows, of size [21,128], are stored in a three-dimensional array (EEG_win) in a .npz file, where the first dimension represents the number of windows per patient. Only recordings containing a seizure interval were considered, and the file names follow the format PatID_seizure_EEGwindow_1.npz, with the number being irrelevant.

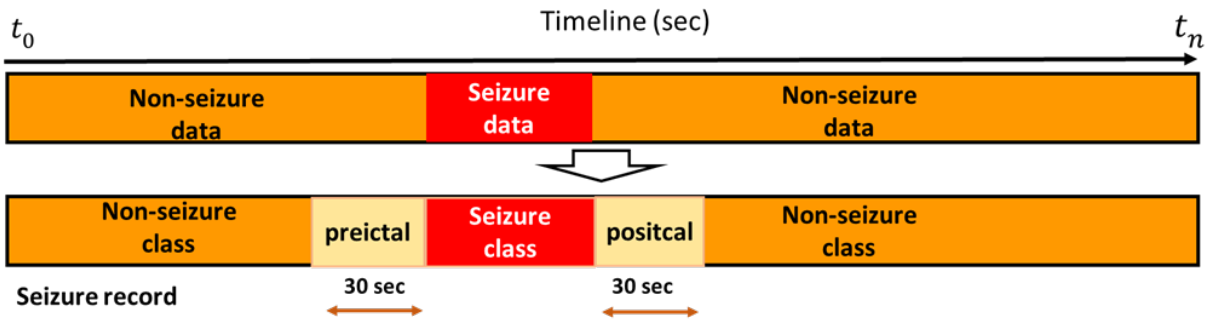

Figure 1. Window Annotation

For each patient, the metadata is stored in parquet format and includes the following fields:

1. class: A binary variable indicating whether a seizure is present in the window (1 for seizure, 0 for non-seizure).
2. filename: The name of the original .edf recording, useful for training and testing patient-specific models.
3. filename_interval: A unique identifier for each seizure episode within a given .edf file. Combined with the filename, it uniquely identifies seizures for a patient.
4. global_interval: A seizure episode identifier that is consistent across all recordings of a patient, ensuring unique seizure tracking.

Figure 2 illustrates an example of this metadata for **PatID = chb01**. Based on the **global_interval** variable, this patient has a total of 7 seizures**,** with 2 occurring in the last recording**, chb01_26.edf**.

	class	filename_interval	global_interval	filename
0	0	1	1	chb01_03.edf
1	0	1	1	chb01_03.edf
2	0	1	1	chb01_03.edf
...	...	...	...	
26525	1	2	7	chb01_26.edf
26526	1	2	7	chb01_26.edf
26527	1	2	7	chb01_26.edf
26528	1	2	7	chb01_26.edf

Figure 2. Metadata example for patient chb01